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Photopigments Underlying Color Vision in Ringtail Lemurs (*Lemur catta*) and Brown Lemurs (*Eulemur fulvus*)

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A recent examination of color vision in the ringtail lemur produced evidence that these prosimians could make color discriminations consistent with a diagnosis of trichromatic color vision. However, it was unclear if this behavior reflected the presence of three classes of cone or whether lemurs might be able to utilize signals from rods in conjunction with those from only two classes of cone. To resolve that issue, spectral sensitivity functions were obtained from ringtail lemurs (*Lemur catta*) and brown lemurs (*Eulemur fulvus*) using a noninvasive electrophysiological procedure, electroretinographic flicker photometry. Results from experiments involving chromatic adaptation indicate that these lemurs routinely have only a single class of cone photopigment in the middle to long wavelengths (peak sensitivity of about 545 nm); they also have a short-wavelength-sensitive cone pigment with peak of about 437 nm. The earlier behavioral results are suggested to have resulted from the ability of lemurs to jointly utilize signals from rods and cones. The cone pigment complements of these lemurs differ distinctly from those seen among the anthropoids.
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Key words: lemurs, cone photopigments, color vision, evolution of photopigments

INTRODUCTION

Although many mammals have at least a minimal color vision capacity, acute trichromatic color vision is found only among the anthropoids [Jacobs, 1993]. The reason for this distinction remains obscure, but as the most primitive of contemporary primates, the prosimians might be expected to provide a key to understanding some aspects of the evolution of primate color vision. That prospect provided one motivation for a study of spectral sensitivity and color vision in the ringtail lemur (*Lemur catta*) carried out in this laboratory [Blakeslee & Jacobs, 1985]. In support of an earlier investigation [Mervis, 1974], we found evidence that these lemurs could make color discriminations. The capacity proved to be far from acute in the sense that (1) although lemurs quickly learned to make brightness discrim-

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inations, extended training was required to yield reliable color discriminations, and (2) even after that point had been reached, the spectral differences required to support color discriminations were consistently large. The nature of color vision in the ringtail lemur was perhaps most compellingly illustrated in a test of color matching. In this test the animals were required to discriminate various additive mixtures of red and green light from a monochromatic yellow light. Two subjects succeeded in making such discriminations for a wide range of red/green mixture proportions under conditions that should have obviated the use of brightness cues, and each animal failed to make the discrimination for some relatively restricted set of mixture proportions. The implication of this result is that the ringtail lemur must have dichromatic color vision in this part of the spectrum. In conjunction with additional evidence for the presence of a third spectral mechanism sensitive to short wavelengths, this suggested that ringtail lemurs are trichromats.

Although the conclusion that lemurs can make trichromatic color discriminations seemed reasonable, it was difficult to infer the nature of the retinal mechanisms supporting the discrimination. The difficulty can be best appreciated from an examination of the earlier color matching results. Shown in Figure 1 are the proportions of red and green light that each of three kinds of primate subjects failed to discriminate from the standard yellow light. Note that the two lemurs failed the discrimination only when the mixture contained a very high proportion of the red component, much higher than that of normal human trichromats equivalently tested, and even somewhat higher than that of human protanomalous trichromats. In evaluating this result (i.e., that the mixture at the match point has a very high proportion of red light) it was noted that it is similar to results obtained from tests done on *dichromatic* humans which had been interpreted as indicating that human dichromats can utilize signals from rods in addition to signals from a single type of cone to make discriminations consistent with trichromacy [Smith & Pokorny, 1977; Nagy, 1980]. Because of this similarity we pointed out that although the ringtail lemur could make trichromatic discriminations, this did not necessarily mean that the animal has three classes of cone [Blakeslee & Jacobs, 1985]. Rather, it might be that ringtail lemurs, like dichromatic humans, are able to exploit signals from rods in some explicit tests of color vision. This was suggested to be particularly likely for animals like lemurs, because with only relatively small numbers of cones it would be difficult to arrange test situations where signals from rods could be eliminated completely.

The presence of three classes of cone photopigment is routine among Old World monkeys. New World monkeys often have only two classes of cone pigment, but there are frequent polymorphic variations allowing three classes of cone pigment in some individuals [Jacobs, 1991]. To which of these patterns, if either, does the lemur conform? The present investigation was intended to answer that question and in so doing better reveal the photopigment basis for spectral discriminations in diurnal lemurs. Over the past several years we have developed a noninvasive electrophysiological technique, electroretinogram (ERG) flicker photometry, that allows a rapid and reliable means to infer the photopigments present in individual retinas [Neitz & Jacobs, 1984; Jacobs & Neitz, 1987]. This technique has been used to measure the pigment complements of two types of diurnal lemur, the ringtail lemur (*Lemur catta*) and the brown lemur (*Eulemur fulvus*).

METHODS

Subjects

The subjects were four, adult ringtail lemurs (three males, one female) and four, adult brown lemurs (three males, one female). The proper taxonomic desig-

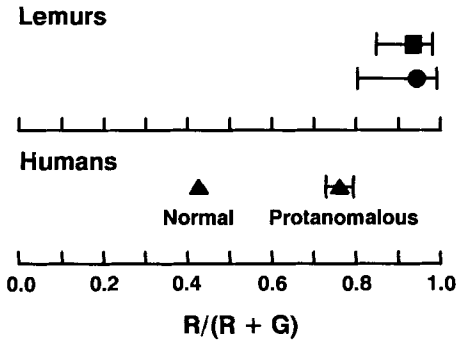


Fig. 1. Results from a test of red/green color matching. In this test subjects were required discriminate various additive mixtures of red and green light (values on the abscissa) from a monochromatic yellow light. The data points show the mixture proportions that the subjects were unable to discriminate (i.e., these values give the color matches). **Top:** Results from two ringtail lemurs. The solid symbols show the midpoint of the matching range; the horizontal bars enclose the entire range of mixtures the animal was unable to discriminate from yellow. **Bottom:** Analogous results for two human subjects; one had normal trichromatic color vision; the other was a protanomalous trichromat. These results are replotted from Blakeslee and Jacobs [1985].

nations for the Lemuridae have recently been matters of discussion [Simons & Rumpler, 1988; Groves & Eaglen, 1988; Tattersall & Koopman, 1989]. We follow Simons and Rumpler [1988] in designating the ringtail lemur as *Lemur catta* and the brown lemur as *Eulemur fulvus*.

Apparatus

The apparatus and procedures employed in ERG flicker photometry have been provided in earlier publications [Neitz & Jacobs, 1984; Jacobs & Neitz, 1987]. Accordingly, only a brief description of each of these follows.

Stimuli were produced using a Maxwellian-view optical system. One beam from this system originated from a Bausch and Lomb high-intensity grating monochromator (half-energy passband = 10 nm). This beam served as the test light of a flicker photometer. Two other beams originated from tungsten-halide lamps; one provided the reference light of the photometer while the other could be used to provide an accessory adaptation light. The three beams were optically combined so as to illuminate a 57° circular spot on the retina. The test and reference beams each contained a high-speed electromagnetic shutter. All three lamps were underrun at 11 V from a regulated DC power supply. Energy measurements were made with a silicon diode photodetector (model PIN 10 DL, United Detector Technology).

ERGs were differentially recorded with a bipolar, contact-lens electrode. In the flicker photometric procedure ERGs elicited by a train of light pulses from the test light are compared to those elicited by a similar train of pulses from the reference light. The two pulse trains are interleaved such that there is a dark interval equal in duration to that of the test and reference lights between successive light pulses. Over presentations of this pulse train the radiance of the test light is varied by adjusting the position of a neutral density wedge in the test light beam. This is done iteratively until the response elicited by the test light is identical in amplitude to that elicited by the reference light; the latter has a constant spectral energy distribution over the course of an experiment. The dependent measure is the radiance of the test light required to accomplish this equation for each of a series of monochromatic test lights. The equation of the responses to test and reference

lights is accomplished electronically by passing the amplified ERG signals through a series of filter stages.

Procedure

Lemurs were anesthetized by IM injection of a mixture of ketamine hydrochloride (20 mg/kg) and acepromazine maleate (0.2 mg/kg). Pupillary dilation was achieved by topical application of atropine sulfate (0.04%) and phenylephredine hydrochloride, and the cornea was anesthetized by topical application of proparacaine hydrochloride (0.5%). A contact lens of the Burian-Allen design was positioned on the test eye. Body temperature was maintained normal during the recording and subsequent recovery sessions with the use of a circulating hot-water heater.

With the exception of one experiment (see below) recording was done in a brightly illuminated room. Flicker photometric equations (i.e., a determination of the test light intensity required to match the response obtained from a standard reference light) were obtained by averaging the last 50 cycles of a total of 70 stimulus cycles. The equation values were read from the density wedge to the nearest 0.01 log unit. Photometric equations for each test/reference pair were determined twice during the recording session and these values were averaged together. These two independently obtained equation values were usually separated by less that 0.05 log unit. Stimulus conditions for each of four separate experiments were as follows:

1. Number of cone types in the middle to long wavelengths. A principal goal was to determine whether lemurs have one or more than one type of photopigment that absorbs light in the middle- to long-wavelength portion of the spectrum. To answer this question we employed a procedure that was used with success earlier [Jacobs & Neitz, 1987]. It is a flicker photometric version of a standard test for photopigment univariance. An ERG flicker photometric equation was made between a middle wavelength test light (540 nm) and a long wavelength (630 nm) reference light. The stimulus pulse rate was 50 Hz. The wavelengths of the two lights and the pulse rate were selected so as to make it unlikely that signals from either rods or short-wavelength-sensitive cones would contribute to the recorded responses. These equations were made as the eye was adapted to either an intense 540 or 630 nm light. These two lights were initially equated for adaptive effectiveness; that is, each adaptation light was separately adjusted in intensity until it elevated the threshold for a 540 nm flickering test light by 0.6 log unit.

2. Spectral sensitivity with high frequency stimulation. Complete spectral sensitivity functions were determined for 50 Hz stimulation. Test wavelengths spanned the range from 440–640 nm in steps of 10 nm. The reference light for this experiment was achromatic (color temperature = 2,540 K; corneal radiance = 43.8 μW).

3. Spectral sensitivity of the short-wavelength cone. Spectral sensitivity functions were obtained with test conditions arranged so as to favor contributions from any short-wavelength-sensitive cones. The photometer pulse rate was set to 25 Hz. To minimize contributions from receptors sensitive to longer wavelengths, the eye was concurrently adapted to an intense long-wavelength light. This was obtained by placing a high-pass filter (50% transmission at 580 nm) in the adaptation beam. To maximize the recording of contributions from short-wavelength-sensitive cones, the reference light was set to 460 nm (corneal radiance = 19.7 μW). Test wavelengths were taken at 10 nm intervals from 430–530 nm.

4. Rod spectral sensitivity. Spectral sensitivity was measured in one lemur

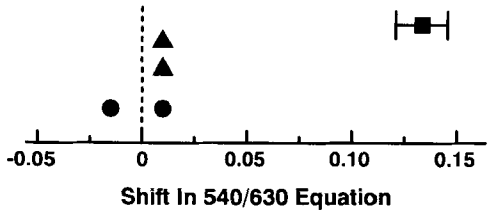


Fig. 2. Results from a test to determine whether lemurs have more than one adaptable photopigment with sensitivity in the spectral range from 540–630 nm. Plotted for each of four lemurs (triangles, brown lemurs; circles, ringtail lemurs) is the difference in the ERG photometric equation for 540 and 630 nm lights as separately determined in the presence of intense 540 and 630 nm adaptation lights. A failure of univariance would lead to a significant positive shift in the equation value. Such a failure is illustrated by the results shown for 15 normal human trichromats who were tested in the same manner (the closed square to the right is the mean shift in the equation for these subjects; the line shows ± 1 SD).

under scotopic test conditions. These conditions included a photometer pulse rate of 15 Hz, a dim, achromatic reference light (corneal radiance = 0.62 μW), an absence of ambient illumination, and a 20 min period of dark adaptation prior to measurement. Spectral sensitivity measurements were made at 10 nm intervals for test lights spanning the range from 440–600 nm.

RESULTS

We found no positive evidence for the presence of more than one middle- to long-wavelength cone type in the lemur, either from the formal chromatic adaptation experiment described above, or in informal tests conducted with several other combinations of 1) type of chromatic adaptation, 2) stimulus pulse rate, and 3) wavelengths of the test and reference lights. The results of the formal test are summarized in Figure 2 which plots the differences in the 540/630 nm equation values determined under the two conditions of chromatic adaptation. A complete absence of a differential chromatic adaptation effect should result in no change in the equation values determined under the two conditions of chromatic adaptation (illustrated by the vertical dashed line of Fig. 2). The differences obtained for four lemurs, two from each genus, each deviate less than 0.02 of a log unit from that prediction. That deviation is within the error of measurement of the ERG flicker photometric technique. Shown for comparison in Figure 2 are the equation changes measured for 15 trichromatic humans tested in the same fashion as were the lemurs. It is of note that these subjects have more than one cone pigment active in the 540/630 nm portion of the spectrum and that this procedure reveals significant chromatic adaptation effects.

The spectral sensitivity functions obtained from each of the eight lemurs are shown in Figure 3. The individual data points represent the average quantal sensitivity values for each test wavelength. The functions obtained for individual animals have been arbitrarily positioned on the sensitivity axis for ease of viewing. There are no obvious differences among the sensitivity functions obtained for individual animals, nor between the animals from the two genera. Each shows peak sensitivity at about 540–550 nm, and there is no systematic indication of any secondary peaks. Further analysis of these spectral sensitivity functions is given below.

To elicit contributions from short-wavelength-sensitive (SWS) cones to the flicker photometric ERG we employed stimulus combinations earlier demonstrated to be effective in recordings carried out on human SWS cones [Crognale et

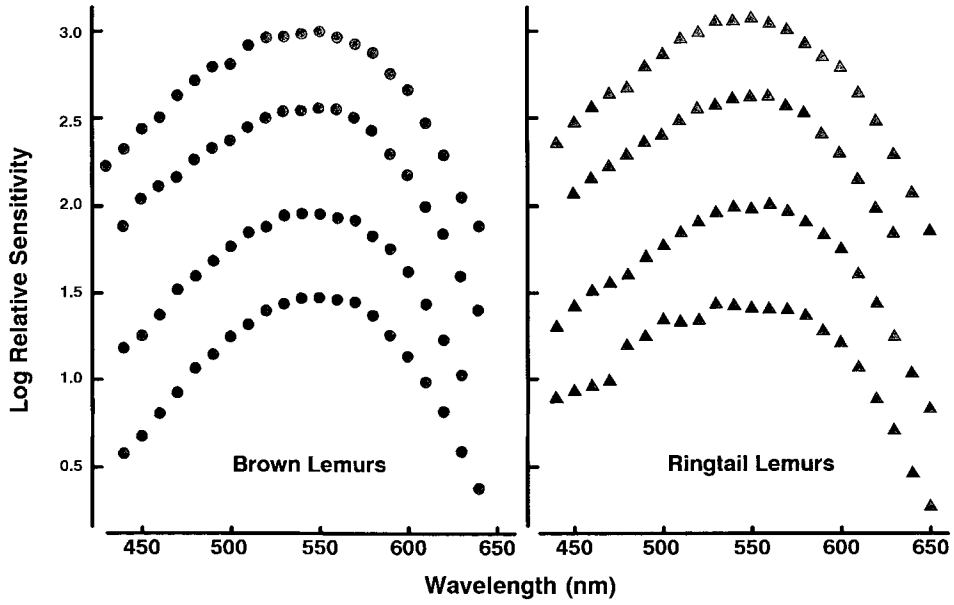


Fig. 3. ERG flicker photometric spectral sensitivity functions for four brown lemurs (**left**) and four ringtail lemurs (**right**). The functions for individual animals have been arbitrarily positioned along the sensitivity axis. Stimulus pulse rate: 50 Hz.

al., 1991]. That this procedure is also effective with the lemur eye is illustrated in Figure 4. Shown there are three spectral sensitivity functions obtained from a brown lemur. One of these (solid circles) was obtained under conditions of neutral adaptation. The other two were obtained when the eye was concurrently illuminated with two different intensities of a long wavelength adaptation light. The three functions have been equated at 520 nm. The dimmer of the two adaptation lights (squares) resulted in a relative increase in sensitivity for test wavelengths shorter than about 510 nm. The brighter of the two adaptation lights (Fig. 4, triangles) yielded a much larger elevation in the short wavelengths. In each case the contribution of the lemur SWS cone to the recorded signal is obvious.

Spectral sensitivity functions obtained from three lemurs (one *E. fulvus*, two *L. catta*) under the brighter of the two adaptation conditions of Figure 4 are shown in Figure 5. The solid circles are mean sensitivity values for the three animals; the vertical lines enclose the total range of sensitivity determined at each test wavelength. Under these conditions, sensitivity is maximal at about 440 nm; it drops from that point out to about 510 nm and then stays effectively constant toward the longer wavelengths. With these test conditions sensitivity was sufficiently low that it became impractical to make photometric equations for test lights of wavelengths greater than about 550 nm.

The scotopic spectral sensitivity function obtained from a dark-adapted ringtail lemur is given in Figure 6. The individual sensitivity values given in Figure 6 have been best fit to the absorption curve for a single photopigment. The spectral positioning of the latter (continuous line) was determined by shifting a standard retinal$_1$ photopigment curve [Ebrey & Honig, 1977] along a log wavenumber axis [Baylor et al., 1987] in incremental steps of 1 nm until a best least-squares fit was

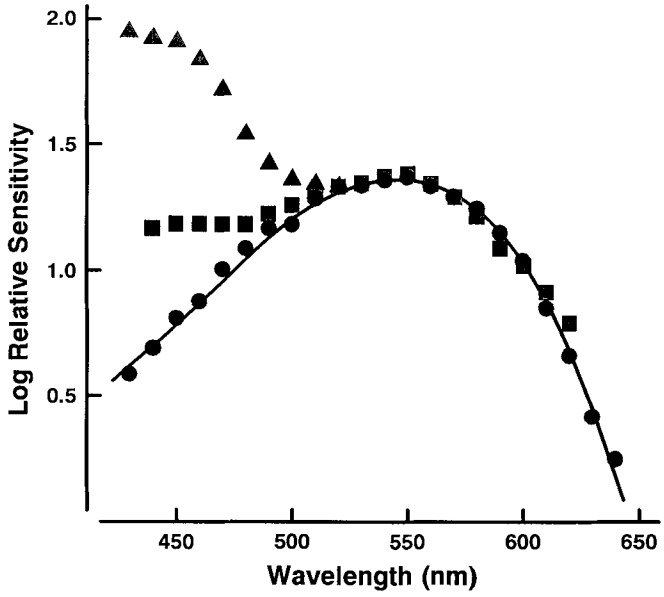


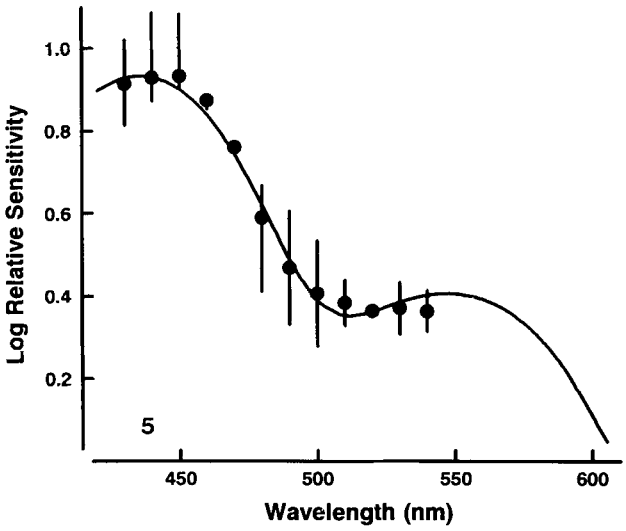
Fig. 4. ERG flicker photometric spectral sensitivity functions obtained from a brown lemur for three different conditions of adaptation: neutral adaptation (solid circles) and for two different intensities of long-wavelength adaptation (squares and triangles). The sensitivity values obtained under the three conditions have been shifted vertically so that they are equal at 520 nm. The continuous line is that for a photopigment having maximum absorption at 545 nm. Stimulus pulse rate: 25 Hz.

obtained between the data array and the pigment absorption curve. The best fit curve so obtained (Fig. 6) has a peak value of 509 nm.

DISCUSSION

Lemur Photopigments

Results of the chromatic-adaptation experiments suggest that under test conditions that employ high-frequency stimulation the lemur eye has only a single adaptable photopigment in the middle to long wavelengths. If that is the case, the spectrum for that pigment should be revealed in the spectral sensitivity functions of Figure 3. Figure 7 plots the averaged spectral sensitivity functions obtained with 50 Hz stimulation for the two types of lemur. In that figure the solid symbols are mean sensitivity values; the vertical bars enclose ± 1 SD. Prior to averaging, the data for individual animals were shifted along the sensitivity axis so that the array of values for each animal had the same average sensitivity. It can be seen that there are only small variations in relative spectral sensitivity among the animals. On the assumption that these sensitivity values represent the operation of a single photopigment, the data arrays were best fit to pigment absorption curves using the procedure outlined above. Those curves (Fig. 7, solid lines) account well for the spectral sensitivity functions. The peaks of the best-fit curves are very similar for brown and ringtail lemurs (544 and 546 nm, respectively). Photopigment absorption curves were best fit to the data from each of the individual subjects as well. Examined in this way, the spectral peaks for the two types of lemur are not significantly different ($t = 1.56$, $df = 6$, $P > .10$). This experiment, in conjunction with the results from chromatic adaptation, indicates that lemurs



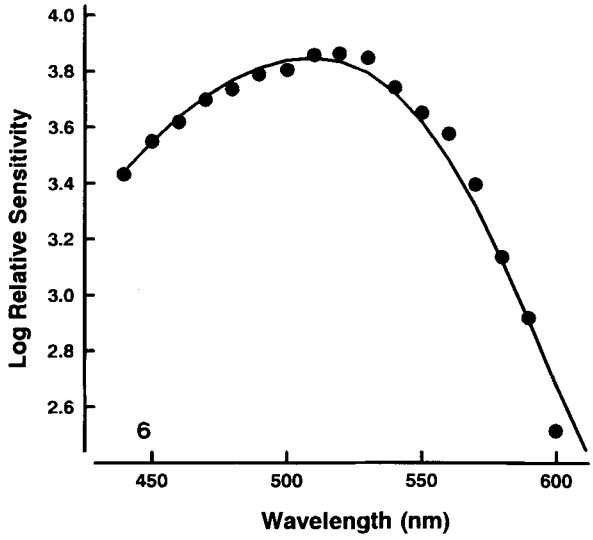


Fig. 5. Spectral sensitivity functions obtained from three lemurs while the eye was concurrently adapted to a bright orange light. The solid circles are mean sensitivity values; the vertical lines enclose the total range of sensitivity for the three animals. The continuous line is the best-fitting summation of two photopigment absorption curves having peaks values of 437 and 545 nm (see text). Stimulus pulse rate: 25 Hz.

Fig. 6. Spectral sensitivity function obtained from a ringtail lemur under conditions of dark adaptation. The continuous line is the curve for the best-fitting photopigment absorption function (peak = 509 nm). Stimulus pulse rate: 15 Hz.

have a single cone photopigment in the middle to long wavelengths; it has a peak sensitivity value of about 545 nm.

With intense chromatic adaptation it proved possible to also record contributions from an SWS cone in the lemur (Fig. 5). However, even with such adaptation it was not possible to isolate the activity of this cone type and thus

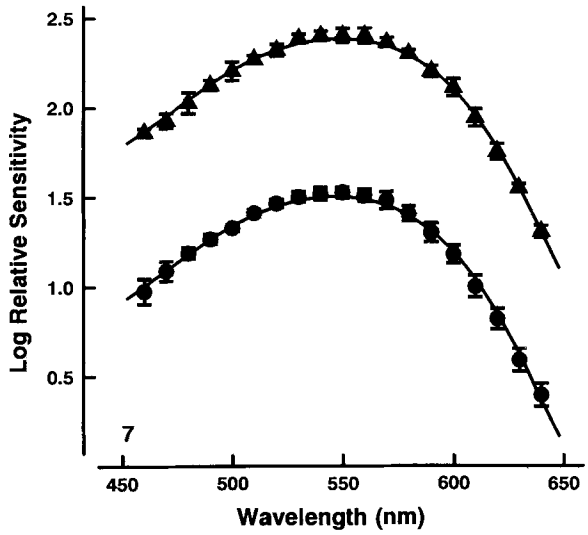


Fig. 7. Mean spectral sensitivity functions ringtail lemurs (top) and brown lemurs (bottom). The solid symbols are mean values for four animals each; the vertical bars enclose ± 1 SD. The continuous lines are the curves for the best-fitting photopigment absorption functions; these have peak values of 546 nm (top) and 544 nm (bottom). Stimulus pulse rate: 50 Hz.

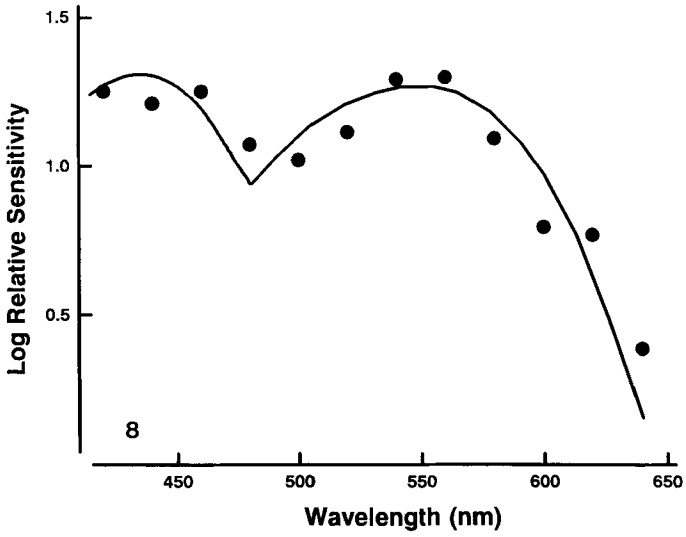


Fig. 8. Spectral sensitivity functions obtained from two lemurs in an increment-threshold test (data from Blakeslee and Jacobs [1985]). The background light for this test was achromatic and had a luminance of 3.4 cd/m^2. The solid symbols are the average sensitivity values for two animals. These data points have been best fitted by summing the sensitivity curves for the two lemur cone pigments (peaks of 437 and 545 nm). The continuous line represents the upper envelope of that best fitting combination.

examine its spectral sensitivity directly. Accordingly, to estimate the spectral location of the lemur SWS cone we assumed that under the test conditions of Figure 5 the ERG indexes the joint activity of the longer cone (Fig. 7) and the SWS

cone, that the outputs from these two cone types are linearly summed to generate the recorded ERG, and that any preretinal absorption over the test wavelengths does not significantly distort the relationship between the pigment spectra and spectral sensitivity. The spectral sensitivity values were best fit to summative combinations of the longer (peak = 545 nm) and various potential SWS cone spectra. To accomplish this we used an analysis program that seeks the best summative fit of two absorption curves. One of these was specified as having a 545 nm peak and the other was sought through an iterative search involving all possible combinations of peak location of an SWS cone (in steps of 1 nm from 420–450 nm) and relative proportions (in increments of 1%) of the two pigment curves. The best fit so determined (437 nm [76%] + 545 nm [24%]) is given by the continuous line in Figure 5. Given the assumptions of fitting procedure, we conclude that the SWS cone of the lemur has peak value in the vicinity of 437 nm. Given that SWS photopigments cannot in general be measured as accurately as other photopigments, the location of the lemur SWS pigment is nevertheless reasonably close to the spectral positions established for a variety of Old and New World monkeys [Jacobs, 1991].

In conjunction with results of the analysis of scotopic spectral sensitivity (Fig. 6), these experiments suggest that lemurs have three classes of photopigment: a rod (peak = c. 509 nm) and two types of cone (peaks of about 437 and 545 nm). To our knowledge there have been no other direct measurements of photopigments for members of either of these genera. Bowmaker [1991] has reported microspectrophotometric (MSP) measurements made on a small number of receptors from the retina of the black lemur (*Eulemur macaco*). Nine rods so measured had an average peak of 501 nm. Spectra were also measured with MSP for three individual cones; these had an average peak value of 543 nm. Given the small number of receptors involved, both of these estimates seem reasonably close to those estimated for electrophysiology. No SWS cones were apparently detected with MSP. That fact is not particularly surprising since SWS cones are likely to be only very sparsely represented in the lemur retina.

In our effort to infer photopigment spectra from ERG measurements we have so far ignored any possible role of tapetal reflection. The presence of a tapetum allows a double passage of light through the photoreceptors and thus any spectrally selective reflection from the tapetum can potentially influence spectral sensitivity. Whereas it is clear that the ringtail lemur has a tapetum [Martin, 1990], it is doubtful that the eye of the brown lemur has such a structure [Wolin & Massopust, 1970]; a tapetum is certainly not obvious in ocular fundus photographs of the brown lemur [Wolin & Massopust, 1967]. Any contribution of tapetal reflection to the measured spectral sensitivities seems to have been small. The data points for values from about 530–580 nm of the scotopic spectral sensitivity function of the ringtail lemur in Figure 6 show a slight elevation above the predicted best-fit curve, and this may indicate the effects of tapetal reflection. On the other hand, the photopic curves show little or no deviations from the absorption curves for either of the types of lemur. We do note for the record that the average spectral peak for the ringtail lemurs was slightly (though not significantly) longer than the comparable value for the brown lemurs. Such a difference would be consistent with the presence of some tapetal reflection in ringtails, but none in brown lemurs. The tapetum is an adaptation to a nocturnal lifestyle, and its role in diurnal species like the ringtail has been questioned [Martin, 1990]. The comparison of spectral sensitivity in lemurs with and without tapeta (Fig. 7) suggests that, whatever else, the tapetum has little or no influence on photopic sensitivity.

Photopigments, Color Vision, and Spectral Sensitivity of Lemurs

In our previous study of lemur color vision we reported that ringtail lemurs can apparently make dichromatic color discriminations in the red/green portion of the spectrum. In consideration of the nature of the color matches these animals made in this part of the spectrum, it was suggested that this behavior might result not from signals from two cone types but rather from the joint use of information from a single cone type and rods. The present experiment makes that interpretation appear likely. The implication is that if it were possible to arrange a test situation where lemur color vision depended entirely on cone-based information, it seems reasonable to expect that lemurs would behave as dichromats and would be unable to make any color discriminations in the middle to long wavelengths. The fact that lemurs were able to use rod and cone signals jointly in a laboratory test raises the intriguing possibility that they may also do so in normal viewing situations.

Lemur spectral sensitivity was measured in earlier behavioral experiments [Blakeslee & Jacobs, 1985]. How well do the photopigments identified in this experiment account for spectral sensitivity? The solid circles in Figure 8 are the average increment-threshold spectral sensitivity measurements obtained from two ringtail lemurs. These were obtained under conditions of light adaptation and thus might be expected to reflect the operation of cones. The continuous line in Figure 8 is the best fit obtained between the absorption curves for the two lemur cone pigments (peaks of 437 and 545 nm, respectively) and the behavioral measurements. The fit was obtained with the assumption that behavioral sensitivity with this measurement condition might follow the upper envelope of the absorption curves of the two cone pigments. With this assumption the behavioral results are well accounted for by the spectra of two cone pigments, and this lends further credence to the conclusion that the lemur eye contains these two classes of photopigment.

Phylogeny of Primate Cone Pigments

All facets of photopic vision in primates are constrained by the number of cone classes, the spectral absorption properties of these cone classes, and several other details of cone representation in the retina. With respect to number of cone classes, there appear to be two predominant patterns among the anthropoids [Jacobs, 1991]. As far as we now know, all the Old World monkeys, apes, and humans are routine trichromats with three well-defined types of cone photopigment. There are much-studied polymorphic variations in the cone complements of a small fraction of the human population, but thus far no similar variation has been detected among other Old World species. With the exception of perhaps only *Aotus,* the New World monkeys show a quite different photopigment pattern [Jacobs, 1993]. Among these genera there are widespread sex-linked polymorphisms in cone pigment complement (and color vision). The modal pattern of New World monkeys is two cone classes with variation in the spectral positioning of one of these cone classes; in addition, as many as two-thirds of all female monkeys have three classes of cone and trichromatic color vision, and they also show polymorphic variation. Consideration of the genetic basis of these variations has led to the suggestion that the pattern of pigments and consequent color vision seen among New World monkeys may be representative of a stage through which primate color vision has evolved to the routine trichromacy of Old World anthropoids. The idea is that the baseline mammalian pattern includes two

classes of cone photopigment, specified respectively by the activity a single autosomal gene (the SWS cone) and a single X-chromosome gene (the longer cone type). A first step toward trichromacy might be the addition of allelic versions of the gene specifying the longer cone type. This provides polymorphic variation in male dichromacy and, since females have two X-chromosomes, females heterozygous for allelic versions of the gene may achieve three classes of cone and trichromatic color vision. To make trichromacy routine requires gene duplication, thus adding a second gene locus on the X-chromosome. It is presently indeterminate whether gene duplication actually followed the divergence of X-chromosome photopigment genes or preceded it, but it seems reasonable to imagine that the two steps represent stages in the evolution of routine trichromatic color vision.

The cone photopigment complements of the two genera of lemurs here examined differ from the patterns established for the anthropoids. Unlike other Old World primates, these prosimians do not have three classes of cone photopigment. The prosimians also differ from the pattern established for many New World monkeys. We think it significant that, although they have two classes of cone photopigment like that of many individual New World monkeys, these lemurs show no evidence of the widespread pigment polymorphism that is a hallmark of many of the platyrrhines. All eight of the lemurs tested had the same middle- to long-wavelength photopigment. This would be a very unlikely result if lemurs had a photopigment polymorphism of the type exemplified by, say, the squirrel monkey [Jacobs & Neitz, 1987]. Indeed, if one wishes to search for resemblances to the photopigment complement of the diurnal lemurs, the closest matches are to be found not among the primates, but rather with members of other nonprimate mammalian genera. For instance, many carnivores that have been examined are like the lemurs in that 1) they have afoveate retinas containing only a modest proportion of cone receptors, 2) they have two classes of cone photopigment and dichromatic color vision, and 3) there is no evidence for any photopigment polymorphism [Jacobs, 1993]. Although we still lack information about many other prosimian genera, it would appear that evolutionary steps toward primate trichromatic color vision have occurred only in the anthropoid radiation.

A final point concerns the spectral positioning of the primate cone pigments. It appears that many of the middle- to long-wavelength cone pigments found in primates occupy one of six spectral positions such that the peaks of these pigments are at stepped locations between about 530 and 560 nm [Jacobs, 1993]. The X-chromosome genes specifying primate cone opsins show high sequence homologies, and it has been suggested that the principal mechanism for evolutionary change among these several possible types is unequal meiotic recombination [Nathans et al., 1986]. There is in fact evidence to suggest that these spectral variations may result from combinations of only three amino acid substitutions in the photopigment molecule [Neitz et al., 1991]. As the number of known photopigment gene sequences increases, it should eventually be possible to determine which among the several possible candidates was the middle- to long-wavelength cone pigment of the earliest primates. From the measurements of primate pigment spectra so far available we suggest that the prototypic cone pigment subserving vision in the middle to long wavelengths may have been one with a peak of about 543–545 nm. Support for that possibility is that not only is this the cone pigment we have identified in these prosimians, but that pigment is also found in a number of the most primitive New World monkeys, including various Callithrichid species and the nocturnal monkey, *Aotus* [Jacobs et al., 1993].

CONCLUSIONS

1. Analysis of the electrical signals generated by retinal cones indicate that members of two genera of lemurs (*Lemur catta* and *Eulemur fulvus*) have identical cone pigment complements. These animals have two classes of cone with respective peak sensitivities of about 437 nm and 545 nm.

2. These results suggest that previous measurements of color discriminations which indicated that lemurs have trichromatic color vision probably reflect the fact that under some test conditions these animals can jointly utilize signals from cones and from rods.

3. In general, anthropoid species either routinely have three classes of cone photopigment or show significant photopigment polymorphisms. The cone pigment complements of ringtail and brown lemurs show neither of these features, and thus their photopigment arrangement is qualitatively more similar to that seen among some other mammals than that characteristic of any of the anthropoids.

4. Comparison of the spectra of the middle- to long-wavelength photopigments found among contemporary primates suggests the possibility that the spectral positioning of the prototypic primate cone pigment may have been like that found in the lemurs examined in this study (i.e., with a spectral peak of about 543–545 nm).

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REFERENCES

Baylor, D.A.; Nunn, B.J.; Schnapf, J.L. Spectral sensitivity of cones of the monkey *Macaca fascicularis*. JOURNAL OF PHYSIOLOGY 390:145–160, 1987.

Blakeslee, B.; Jacobs, G.H. Color vision in the ring-tailed lemur (*Lemur catta*). BRAIN, BEHAVIOR AND EVOLUTION 26:154–166, 1985.

Bowmaker, J.K. Visual pigments and colour vision in primates. Pp. 1–9 in FROM PIGMENTS TO PERCEPTION: ADVANCES IN UNDERSTANDING VISUAL PROCESSES. A. Valberg, B.B. Lee, eds. New York, Plenum, 1991.

Crognale, M.A.; Jacobs, G.H.; Neitz, J. Flicker photometric ERG measurements of short wavelength sensitive cones. DOCUMENTA OPHTHALMOLOGICA PROCEEDINGS SERIES 10:341–346, 1991.

Ebrey, T.G.; Honig, B. New wavelength-dependent visual pigment nomograms. VISION RESEARCH 17:147–151, 1977.

Groves, C.P.; Eaglen, R.H. Systematics of the Lemuridae (Primates, Strepsirhini). JOURNAL OF HUMAN EVOLUTION 17: 513–538, 1988.

Jacobs, G.H. Variations in colour vision in nonhuman primates. Pp. 199–214 in INHERITED AND ACQUIRED COLOUR VISION DEFICIENCIES: FUNDAMENTAL ASPECTS AND CLINICAL STUDIES. D.H. Foster, ed. London, Macmillan, 1991.

Jacobs, G.H. The distribution and nature of color vision among the mammals. BIOLOGICAL REVIEWS, 1993 (in press).

Jacobs, G.H.; Neitz, J. Inheritance of color vision in a New World monkey (*Saimiri sciureus*). PROCEEDINGS OF THE NATIONAL ACADEMY OF SCIENCES USA 84:2545–2549, 1987.

Jacobs, G.H.; Deegan II, J.F.; Neitz, J.; Crognale, M.A.; Neitz, M. Photopigments and color vision in the nocturnal monkey, *Aotus*. VISION RESEARCH, 1993 (in press).

Martin, R.D. PRIMATE ORIGINS AND EVOLUTION. Princeton, Princeton University Press, 1990.

Mervis, R.F. Evidence of color vision in a diurnal prosimian, *Lemur catta*. ANIMAL LEARNING AND BEHAVIOR 2:238–240, 1974.

Nagy, A.L. Large-field substitution Rayleigh matches of dichromats. JOURNAL OF THE OPTICAL SOCIETY OF AMERICA A 70:778–784, 1980.

Nathans, J.; Thomas, D.; Hogness, D.S. Molecular genetics of human color vision: The genes encoding blue, green and red pigments. SCIENCE 232:193–202, 1986.

Neitz, J.; Jacobs, G.H. Electroretinogram measurements of cone spectral sensitivity

in dichromatic monkeys. JOURNAL OF THE OPTICAL SOCIETY OF AMERICA A 1:1175–1180, 1984.

Neitz, M.; Neitz, J.; Jacobs, G.H. Spectral tuning of pigments underlying red-green color vision. SCIENCE 252:971–974, 1991.

Simons, E.L.; Rumpler, Y. *Eulemur:* New generic name for species of *Lemur* other than *Lemur catta.* COMPTES RENDUS DE L'ACADEMIE DES SCIENCES PARIS SERIE III 307:547–541, 1988.

Smith, V.C.; Pokorny, J. Large-field trichromacy in protanopes and deuteranopes. JOURNAL OF THE OPTICAL SOCIETY OF AMERICA A 67:213–220, 1977.

Tattersall, I.; Koopman, K. A further note on nomenclature in Lemuridae. JOURNAL OF HUMAN EVOLUTION 18:499–500, 1989.

Wolin, L.R.; Massopust, L.C., Jr. Characteristics of the ocular fundus in primates. JOURNAL OF ANATOMY 101:693–699, 1967.

Wolin, L.R.; Massopust, L.C., Jr. Morphology of the primate retina. Pp. 1–27. In THE PRIMATE BRAIN. C.R. Noback and W. Montagna, eds. Appleton-Century-Crofts, New York, 1970.